Aman Jayesh

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Education

International Institute of Information Technology, Hyderabad **Telangana**

Bachelor of Technology in Computer Science and Engineering *July 2024 – May 2028*

- **Relevant Coursework**: Introduction to Information Security, Software Programming for Performance, Operating Systems and Networks, Data Structures and Algorithms, Computer Systems Organization, Introduction to Software Systems, Embedded Systems Workshop, Probability & Statistics.

Research

Undergraduate Researcher at Center for Security, Theory and Research **IIIT Hyderabad**

Under Prof. Ankit Gangwal *March 2026 – Present*

- Working on ML security topics including backdoor attacks and adversarial examples, with a keen interest in federated learning security.

Projects

SAMSTeG: Noise-Robust Keyword Spotting | *Deep Learning, TinyML* [GitHub] CVEST Lab IIITH

- Authored a research letter for **IEEE Embedded Systems Letters Journal** (still under review) detailing a novel noise-robust keyword spotting architecture for extreme edge devices under Prof. Abhishek Srivastava.
- Proposed **Multi-Scale Temporal Gating (MSTG)** and **Hybrid Dropout** and achieved **95.71%** accuracy on Google Speech Commands, matching SOTA models like **BC-ResNet**.

Respire | *Deep Learning, Signal Processing* CVEST Lab IIITH

- Worked on lung sound classification for the **Respire** paper at **CVEST** with PhD and Masters students to build models for phones and ESP32.
- Utilized signal processing techniques (Gammatone, Mel spectrograms) and obtained **82%** and **87%** accuracies on wheeze and crackle classes respectively on the HF-Lung dataset.

OSCAPegoat | *Python, LLMs* [GitHub] CyberMANTHAN Lab IIITH

- Researched benchmarking AI-driven system security compatibility based on OpenSCAP STIG analysis at **CyberManthan IIITH**.
- Benchmarked how local LLMs fare at remedying failing OSCAP tests across **5 different models** using scripts with heavy error handling.
- Achieved up to **70% remediation success** via a **3-iteration self-correcting loop**, enabling fully automated compliance fixing.

OpenWheel-Gnome | *Linux, C++* [GitHub]

- Developed a Linux driver for the ASUS ProArt dial featuring seamless GNOME desktop support.
- Engineered a user-friendly graphical interface for option switching and custom profile creation.

OpenClaw Vulnerability Investigation | *Cybersecurity, Shodan, OSINT*

- Conducted a thorough investigation into vulnerable OpenClaw (formerly ClawdBot) instances in India using **Shodan**.
- Compiled a comprehensive registry of exposed instances, including geolocation data and detailed system information.
- Analyzed the security vulnerabilities associated with these exposures and documented the potential attack vectors.

Achievements & Extracurriculars

- Kerala State Team Chess Champion.
- Team Head at Open Source Developers Group (OSDG), IIIT Hyderabad.

Technical Skills

Languages: C, C++, Python
Libraries & Frameworks: TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib, Librosa
Tools & Platforms: Git, GitHub, Shodan, ESP32, Edge Impulse, HuggingFace, Ollama
Certifications: Machine Learning Specialization (DeepLearning.AI – Stanford)